1 The Basics

We will deal with smooth (Riemannian) manifolds throughout. Some common notation: (M, g) is a manifold dimension n, (N, h) a manifold dimension k, $p \in U \subseteq M$ open, $q \in V \subseteq N$ open, $F : M \to N$ smooth with $F(p) = q$.

There is still a theorem for showing something is a manifold without introducing the technical definition.

Theorem 1. *Let M be a set and suppose we are given a collection $\{U_\alpha\}_{\alpha \in A}$ of subsets of M and a collection of injective maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ for each α such that the following hold:*

- *for each α the set $\tilde{U}_\alpha = \varphi_\alpha(U_\alpha)$ is an open subset of $\mathbb{R}^n$,*
- *for all α, β the set $\varphi_\beta(U_\alpha \cap U_\beta)$ is an open subset of $\mathbb{R}^n$,*
- *for all α, β the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is a homeomorphism/smooth,*
- *whenever $x, y \in M$ are two distinct points then either there exists α with $x, y \in U_\alpha$ or there exists α, β with $x \in U_\alpha, y \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$.*
- *there is a countable $I \subset A$ such that the collection $\{U_\alpha\}_{\alpha \in I}$ covers M.*

Then M has a unique topology with respect to which the collection $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is an atlas for M (hence M is a topological / smooth manifold).

Definition 1. The **tangent space** $T_p\mathbb{R}^n$ is a n-dimensional vector space with basis $\{e_i|_p\}_{i=1}^n$, where $e_i|_p : C^\infty(M) \to \mathbb{R} : e_i(f) = \frac{\partial f}{\partial x_i}\Big|_p$. The **tangent space** T_pM is spanned by the basis $\{\partial_i|_p\}_{i=1}^n$, where $\partial_i|_p = d\phi^{-1}|_{\phi(p)} e_i|_{\phi(p)}$.

I should have define $d\phi^{-1}$ before, but the following will almost do:

Definition 2. dF_p is a linear function $T_pM \to T_qN$ defined by specifying where basis get sent: given

$$\{\partial_1|_p, \partial_2|_p, \cdots \partial_n|_p\}, \{\widetilde{\partial}_1|_q, \widetilde{\partial}_2|_q, \cdots \widetilde{\partial}_k|_q\}$$

basis of T_pM and T_qN respectively, define the action by the matrix

$$\begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \\ \frac{\partial F_k}{\partial x_1} & & \frac{\partial F_k}{\partial x_n} \end{bmatrix}$$

Suppressing $p \in M$ gives you a vector field:

Definition 3. A **vector field** X is a map $M \to TM$ fibered over $p \in M$. Using Einstein Summation, we can write $X = X^i \partial_i$ where X_i are $M \to \mathbb{R}$ functions. Collection of all vector fields are denoted $\Gamma(TM)$.

Another way is to see $X : C^\infty(M) \to C^\infty(M)$, writing $Xf = X_i \frac{\partial f}{\partial x^i}$ (note that with p added, they are pointwise multiplication). Note that it satisfy **chain rule**: $X(fg) = fXg + gXf$.

Note that $\{\partial_i|_p\}$ forms a basis for all $p \in U$. This is also called a **local frame**.

Definition 4. If there are vector fields $\{E_i\}_{i=1}^n$ (maybe defined near an open set $U \subseteq M$) such that $\{E_i|_p\}$ is a basis for T_pM for all $p \in U$, then it is called a **local frame**.

Definition 5.

- F is an **immersion** if dF_p is injective for all $p \in M$.
- F is a **submersion** if dF_p is surjective for all $p \in M$.
- (Actually the Inverse function theorem) if dF_p is bijective, then it is a **local diffeomorphism**.

Definition 6.

- A point p is called a **regular point** if $dF_p : T_pM \to T_pN$ is surjective, otherwise is called a **critical point**.
- A point q is called a **regular value** if every point in $F^{-1}(q)$ is a **regular point**, and it is called a **critical value** otherwise.

Theorem 2. *If q is a regular value, then $F^{-1}(q)$ is an embedded submanifold of dimension $n - k$.*

2 Things, and Their Pullbacks

Now we want to "transport" objects through a smooth function $F : M \to N$, they are (almost) always called "**pullbacks**" because it transport objects from N to M. (The reverse is called **pushforwards**.)

Definition 7. Let M and N be smooth manifolds and F a diffeomorphism. Let X, Y be vector fields on M, N respectively. The **pullback** of Y is the vector field $F^*Y : M \to TM$ on M defined by$F^*(Y)_p = dF^{-1}_{F(p)}\left(Y_{F(p)}\right)$. The **pushforward** of X is the vector field $F_*X : N \to TN$ on N defined by$F_*(X)_q = dF_{F^{-1}(q)}\left(X_{F^{-1}(q)}\right)$

Definition 8. Given a vector space V, an **(alternate) k-forms** on V, denoted as a collection as $\text{Alt}^k(V)$, or more commonly $\Lambda^k(V^*)$, are linear functionals $V \times V \cdots V \to \mathbb{R}$ (k times in the product) such that they are multilinear and alternating.

Note that $\Lambda^1(V^*) = V^*$, and it comes with a wedge product $\wedge$ that produce a $(k+\ell)$-form between a k-form and a ℓ-form.

Definition 9. Let $L : V \to W$ be a linear map, then the **pullback** of $\phi \in \text{Alt}^k(W)$ is defined by $L^*\phi(x_1, x_2, \cdots x_k) := \phi(L(x_1), L(x_2), \cdots L(x_k))$, which is an k-form on V.

Definition 10. A **differential k-form** is a function $M \to \text{Alt}^k(TM)$ fibered over $p \in M$. Denote the collection of differential k-forms on M as $\omega \in \Omega^k(M)$.

That is, ω_p is an alternating k-form on T_pM. Suppressing p, the coefficients are functions (just like vector fields). For example, in $\mathbb{R}^n$ (with standard coordinate), an n-form is in the form of $f dx^1 \wedge dx^2 \cdots \wedge dx^n$, where $dx^i := (e_i)^*$. For integrating forms and actual computation, the following will help:

Definition 11. $\omega \in \Omega^k(N)$ can be **pullback** (by F) to $\Omega^k(M)$ by the formula: $(F^*\omega)_p := (dF)^*_p(\omega_{F(p)})$

Definition 12 (Informal)**.** For any $\omega \in \Omega^n(M)$, we can define (by chart) $\int_M \omega := \int_{\phi(M)}(\phi^{-1})\omega$ noting the following caveats:

- Integrating in $\Omega^n(\mathbb{R}^n)$ is simply the normal definition you would expect.
- It is okay if $\phi(M)$ captures M outside a set of measure 0.

Expressing in coordinates, if $\omega = f dx^{i_1} \wedge dx^{i_2} \cdots \wedge dx^{i_k}$, then

- $df = \frac{\partial f}{\partial x_i}\, dx^i$.
- $d(f dx^{i_1} \wedge dx^{i_2} \cdots dx^{i_k}) = df \wedge dx^{i_1} \wedge dx^{i_2} \cdots dx^{i_k}$.
- $F^*\omega = (f \circ F) dF_{i_1} \wedge dF_{i_2} \cdots \wedge dF_{i_k}$.

Theorem 3. $\int_{\partial M} \omega = \int_M d\omega$ *if* ω *is compactly supported.*

3 Now with the Metric

Definition 13. A **riemannian metric** is an inner product of tangent vectors fibered over p. That is, $g_p : T_pM \times T_pM \to \mathbb{R}$ is an inner product. There is also a smoothness requirement, requiring that $g_{ij}(p) = g_p(\partial_i|_p, \partial_j|_p)$ to be smooth ($M \to \mathbb{R}$ sense)

Example: on $\mathbb{R}^n$, we have the normal dot product $g_{Euc\,p}(e_i|_p, e_j|_p) = \delta_{ij}$.(Kronecker Delta). WARNING: we generally don't have $g_p(\partial_i|_p, \partial_j|_p) = \delta_{ij}$, if I have, it is called an **orthonormal**.

Definition 14. A local frame $\{E_i\}_{i=1}^n$ is **orthonormal** if $g_p(E_i|_p, E_j|_p) = \delta_{ij}$ for any $p \in U$.

An orthonormal frame always exists by the Gram-Schmidt process on any local frame. Remember we can write $\partial_i = A_i^j E_j$ for some matrix $A_i^j|_p$.

Definition 15. Given two riemannian manifold $(M, g), (N, h)$ we can **pullback** a metric by

$$(F^*h)_p(X_p, Y_p) = h_{F(p)}(dF_pX_p, dF_pY_p)$$

, this requires F to be an immersion.

Definition 16. Two manifolds are isometric if there exists diffeomorphism F such that $F^*h = g$. If this happens locally (see homework 3), it is an **local isometry**.

Local isometries does *not* form an equivalence relation but isometries do.

Theorem 4. *Every smooth manifold is a riemannian manifold.*

Since you can "patches up" $\phi^* g_{Euc}$ locally by charts. But one more common construction is to use the inclusion $i : M \to \mathbb{R}^m$ and use $i^* g_{Euc}$ as the metric on M.

Theorem 5. *There exists a unique smooth differential n-form $\omega \in \Omega^n(M)$ such that $\omega(E_1, \cdots, E_n) = 1$ for any oriented local orthonormal frame $(E_1, \cdots, E_n)$ on M.*

There is a way to construct this. Simply take $(E_1)^* \wedge (E_2)^* \cdots \wedge (E_n)^*$, where $\{E_i\}_{i=1}^n$ are any orthonormal frame. A coordinate version of the riemannian volume form is $\omega = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots dx^n$.

Definition 17. A **linear connection** (in TM) is a map $\Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$, denoted by $(X, Y) \mapsto \nabla_X Y$ satisfying the following properties.

1. $\nabla_X Y$ is linear in X over $C^\infty(M)$: $\nabla_{fX_1 + gX_2} Y = f \nabla_{X_1} Y + g \nabla_{X_2} Y, \quad \forall f, g \in C^\infty(M)$.
2. $\nabla_X Y$ is linear in Y over $\mathbb{R}$: $\nabla_X (aY_1 + bY_2) = a \nabla_X Y_1 + b \nabla_X Y_2, \quad \forall a, b \in \mathbb{R}$.
3. ∇ satisfies the following **product rule**: $\nabla_X (fY) = f \nabla_X Y + (Xf) Y, \quad \forall f \in C^\infty(M)$.

Theorem 6. *A connection takes the following coordinate form: $\nabla_X Y = X^i \partial_i Y^j \partial_j + X^i Y^j \Gamma_{ij}^k \partial_k$, where $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$ are called the **Christoffel symbols**. That is, any connection is uniquely determined by the (arbitrary) choice of n^3 $M \to \mathbb{R}$ functions (as Christoffel symbols).*

Definition 18. If a connection satisfies the further 2 conditions:

4. ∇ is symmetric: $\nabla_X Y - \nabla_Y X = [X, Y], \quad X, Y \in \Gamma(TM)$, (Note: $\nabla_X Y - \nabla_Y X - [X, Y]$ is sometimes referred to as the **torsion** of ∇. In this requirement we require ∇ to be torsion-free.)
5. ∇ is compatible with the Riemannian metric g: $Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$

Then ∇ is called a (the) **Levi-Civita Connection**.

Theorem 7. *Levi-Civita Connection is unique on M, this is because the formula $\Gamma_{ij}^k = \frac{1}{2} g^{mk} (\partial_i g_{jm} + \partial_j g_{im} - \partial_m g_{ij})$, where g^{ij} are the inverse of the matrix g_{ij} (at each p) gives us a unique definition of the connection.*

4 Curves and Normal Coordinates

Now we look at $\gamma : I \to M$ curves. I is always an open interval in $\mathbb{R}$. Again we fix a chart so $\phi \circ \gamma(t) := (\gamma^1(t), \gamma^2(t), \cdots \gamma^n(t))$ as $I \to \mathbb{R}$ functions, so $\dot{\gamma}^i$ make sense.

Definition 19. A **vector field along** γ is a smooth function $V : I \to TM$ such that $V(t) \in T_{\gamma(t)} M$. The collection is denoted as $\mathcal{J}(\gamma)$. A $V \in \mathcal{J}(\gamma)$ is **extendable** if there is $\widetilde{V} \in \Gamma(TM)$ such that $\widetilde{V}|_{\gamma(t)} = V(t)$ for all $t \in I$.

Similarly, a vector field along γ V can always be written in coordinate as $V = (V^1, V^2, \cdots V^n)$.

Definition 20. Given γ, we can always define the **velocity field** $\dot{\gamma} := \dot{\gamma}^i \partial_i|_{\gamma(t)}$ in $\mathcal{J}(\gamma)$.

Definition 21. Given ∇ and γ, a map $D_t : \mathcal{T}(\gamma) \to \mathcal{T}(\gamma)$ and satisfies the following properties.

1. $D_t(aX + bY) = a D_t X + b D_t Y, \quad \forall a, b \in \mathbb{R}$.
2. $D_t(fX) = \dot{f} X + f D_t X, \quad \forall f \in C^\infty(I)$.
3. If $X \in \mathcal{T}(\gamma)$ is extendable to $\tilde{X} \in \Gamma(TM)$. Then $D_t X = \nabla_{\dot{\gamma}} \tilde{X}|_\gamma$.

This takes a coordinate expression $D_t V = (\dot{V}^k + \dot{V}^i \dot{\gamma}^j \Gamma_{ij}^k) \partial_k$. (Hence ∇ is given.)

Definition 22. A curve (subject to ∇) is a **geodesic** if $D_t \dot{\gamma} \equiv 0$.

Definition 23. The equation $(\ddot{\gamma}^k + \dot{\gamma}^i \dot{\gamma}^j \Gamma_{ij}^k) = 0$ is called the **geodesic equation**.

Theorem 8 (Existence and Uniqueness of Geodesic)**.** *For any $(p, v) \in TM$ and any $t_0 \in I \subset \mathbb{R}$, there exist a geodesic $\gamma : I \to M$ satisfying the initial conditions $\gamma(t_0) = p, \dot{\gamma}(t_0) = v$. Moreover, if $\gamma_1 : J \to M$ is another geodesic such that $t_0 \in J$ and satisfies the geodesic equation, then $\gamma(I \cap J) = \gamma_1(I \cap J)$.*

Definition 24. A geodesic $\gamma : I \to M$ is said to be **maximal** if there is no geodesic $\tilde{\gamma} : J \to \mathbb{R}$ so that $I \subset J$ and $\tilde{\gamma}(I) = \gamma$.

Corollary. *For any $(p, v) \in TM$, there exists a unique maximal geodesic $\gamma_{(p,v)} : I \to M$ so that $0 \in I$ and $\gamma(0) = p, \dot{\gamma}(0) = v$.*

The maximal geodesics exhibits a nice property with "velocity scaling":

Theorem 9 (Rescaling Lemma)**.** *Let γ be the maximal geodesic above. Then for every $c, t \in \mathbb{R}$, $\gamma_v(ct) = \gamma_{cv}(t)$ when at least one of the two sides is defined.*

We use this to build a chart such that the coordinate gives a nice description of geodesics.

Definition 25. Let $\gamma_{(p,v)}$ be as above. Define $\mathcal{U} = \{(p, v) \in TM : \gamma_{(p,v)} \text{ is defined on } [0, 1]\}$. The **exponential map** $\exp : \mathcal{U} \to M$ is given by $\exp(v) = \gamma_{(p,v)}(1)$.

For each $p \in M$, the **restricted exponential map** $\exp_p$ is the restriction of $\exp$ to $\mathcal{U}_p := \mathcal{U} \cap T_pM$.

Theorem 10 (Properties of $\exp$)**.** *$\mathcal{U}$ is an open subset of TM and $\exp : \mathcal{U} \to M$ is smooth. Moreover,*

- *$\mathcal{U}$ contains $0 \in T_pM$ for every $p \in M$, and for every $p \in M$, $\mathcal{U}_p$ is star-shaped with respect to $0 \in T_pM$.*
- *For every $v \in \mathcal{U}_p$, the geodesic γ_v is given by $\gamma_v(t) = \exp(tv) = \exp_p(tv)$ for all t such that either side is defined.*
- *For each point $p \in M$, the differential $d\left(\exp_p\right)_0 : T_0\left(T_pM\right) \cong T_pM \to T_pM$ is the identity map under the natural identification of T_pM and $T_0\left(T_pM\right)$.*

Then by inverse function theorem on the last point, we immediately have the following definition.

Definition 26. Fix $p \in M$, then there is an open set $V \subset T_pM$ such that $\exp_p|_V$ is an diffeomorphism. The image of this map is a open subset $U \subseteq M$. Now V can be identified with $\mathbb{R}^n$ with an isometry $F : T_pM \to \mathbb{R}^n$ by choosing an orthonormal basis $\{E_i|_p\}$ at p (and sending them to basis $\{e_i\}$ in $\mathbb{R}^n$). So $F^{-1} \circ (\exp_p)^{-1}$ gives a coordinate chart, and it is called the **normal coordinate** for M.